Lecture 10 recap

In Lecture 10:

- Introduced improper integrals and learnt how to assess their convergence
- Defined the Laplace transform and discussed its linear properties
- Set out criteria for existence of the Laplace transform of a function

Lecture 11

In Lecture 11, we will:

1. Investigate existence of the Laplace transform for example functions
2. Introduce the inverse Laplace transform
3. Use the LT "cheat sheet" to calculate LT and ILT in examples

Exercise 3.6: Does the below function have a Laplace transform?

$$f = \begin{cases} 0 & 0 \leqslant t < 1 \\ 1 & 1 \leqslant t \end{cases}$$

Exercise 3.7: Do Laplace transforms exist for the following functions?

$$f_1(t) = t^3\,\mathrm{e}^{2t}; \qquad f_2(t) = \mathrm{e}^{t^2}$$

Inverse Laplace transform

Theorem

Let f be a *continuous* function on $t \in [0, \infty)$ that has a Laplace transform, F. Then the inverse Laplace transform also exists, i.e.,

$$\mathscr{L}^{-1}(F) = f \quad \text{for all } t \in [0, \infty)$$

The inverse Laplace transform is also linear:

1. $\mathscr{L}^{-1}(F + G) = \mathscr{L}^{-1}(F) + \mathscr{L}^{-1}(G) = f + g$
2. $\mathscr{L}^{-1}(\beta F) = \beta \mathscr{L}^{-1}(F) = \beta f$ where β is a constant

*A formula for the inverse exists but it requires advanced complex analysis.

Discontinuous functions

1. If f is discontinuous at $t = 0$, the inverse returns the limit from above, i.e.,

$$\mathscr{L}^{-1}(F)\Big|_{t=0} = \lim_{t \to 0^+} f(t)$$

2. If f is discontinuous at $\tau > 0$, the inverse returns the mid-point

$$\mathscr{L}^{-1}(F)\Big|_{t=\tau} = \frac{1}{2}\left(\lim_{t \to \tau^-} f(t) + \lim_{t \to \tau^+} f(t)\right)$$

MAST20030 Differential Equations Formulae Sheet

1) Laplace Transforms

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$ (Definition of Transform)
2.	1	$\frac{1}{s}$
3.	t^n	$\frac{n!}{s^{n+1}}$
4.	e^{at}	$\frac{1}{s-a}$
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$

Exercise 3.8: Find the following Laplace transforms

(i) $2e^{-3t}$ (ii) $\sin(2t) + \cos(2t)$

Exercise 3.9: Find the following inverse Laplace transforms

$$\text{(i)}\ \frac{2}{s^4} \qquad \text{(ii)}\ \frac{2}{s-1} + \frac{3s}{s^2+9}$$

Partial fractions; **Exercise 3.10**: Find the inverse Laplace transform of

$$F(s) = \frac{2}{(s-1)(s-2)}$$